

WOLF POINT 29 ENE, MT, USA

WMO: 742320

Lat: 48.3082N

Lon: 105.1018W

Elev: 2085

StdP: 13.62

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 94060

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.0	(c) -17.4	(d) -28.6	(e) 1.2	(f) -23.1	(g) -22.7	(h) 1.7	(i) -17.0	(j) 24.6	(k) 29.2	(l) 22.3	(m) 26.5	(n) 3.3	(o) N/A	(p) 0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 29.8	(c) 94.1	(d) 64.6	(e) 90.3	(f) 64.8	(g) 86.6	(h) 63.6	(i) 69.7	(j) 83.1	(k) 67.5	(l) 81.7	(m) 65.8	(n) 80.7	(o) 8.0	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	100.8	77.2	62.7	92.0	73.6	60.5	85.0	71.2	35.1	82.8	33.0	81.9	31.6	80.1	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.3	17.5	15.3		DB	-33.4	100.5	7.7	4.2	-39.0	103.6	-43.5	106.1	-47.8	108.5	-53.5	111.5
				WB	-34.8	74.3	5.4	3.5	-38.7	76.9	-41.9	78.9	-44.9	80.9	-48.8	83.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	13.2	13.7	28.4	41.9	52.3	62.0	69.7	67.2	56.7	42.0	27.4	14.6
	DBStd	22.98	15.51	14.69	14.03	9.64	8.57	6.56	6.10	6.82	8.74	9.86	12.29	14.14
	HDD50	5269	1140	1015	671	272	75	3	0	0	33	282	679	1099
	HDD65	9157	1605	1435	1133	694	402	131	22	57	273	712	1129	1564
	CDD50	1954	0	0	3	28	146	362	611	532	235	36	1	0
	CDD65	366	0	0	0	0	8	40	168	124	25	1	0	0
	CDH74	6410	0	0	2	35	244	796	2570	2052	659	52	0	0
	CDH80	2711	0	0	0	5	54	244	1189	941	269	9	0	0
	WSAvg	6.2	6.0	5.8	6.8	7.5	7.4	6.6	5.8	5.7	5.8	6.2	5.7	5.6
	PrecAvg	13.7	0.4	0.3	0.5	1.0	2.1	3.0	2.2	1.6	0.9	0.9	0.4	0.4
Precipitation	PrecMax	20.4	1.2	0.7	1.6	2.2	5.9	6.1	8.0	6.2	2.5	3.1	1.7	0.9
	PrecMin	8.7	0.1	0.1	0.1	0.4	0.6	0.9	0.6	0.0	0.2	0.1	0.0	0.0
	PrecStd	2.4	0.3	0.2	0.3	0.5	1.4	1.2	1.4	1.3	0.7	0.7	0.4	0.2
	PrecStd	2.4	0.3	0.2	0.3	0.5	1.4	1.2	1.4	1.3	0.7	0.7	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	51.4	70.2	79.1	85.8	92.1	99.9	99.5	93.7	80.4	63.9	48.5
		MCWB	39.6	41.1	50.2	55.8	60.2	66.0	64.1	64.7	63.8	58.4	49.3	38.9
	2%	DB	42.3	43.2	61.3	71.1	80.2	86.4	94.8	93.7	87.5	72.5	56.0	41.6
		MCWB	36.7	36.4	46.6	51.6	58.5	64.3	65.3	63.7	60.4	55.4	44.9	35.6
	5%	DB	38.3	38.0	54.3	66.0	76.2	82.1	90.8	89.6	82.1	65.5	50.4	37.8
		MCWB	34.1	33.9	42.9	48.3	56.7	62.6	65.9	63.5	59.0	50.3	41.5	33.5
	10%	DB	34.7	34.0	48.1	61.1	70.9	78.5	87.0	85.3	76.0	59.7	44.7	34.0
		MCWB	31.6	31.4	40.0	45.8	54.1	60.8	65.5	62.9	57.1	48.0	38.2	31.1
	0.4%	WB	40.2	41.2	50.4	56.0	64.4	72.6	74.8	70.0	68.3	59.5	50.0	39.9
		MCDB	47.4	49.7	67.5	76.0	77.7	83.9	87.9	85.4	84.0	76.5	63.4	47.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.9	36.9	47.5	51.3	61.2	67.1	71.1	67.7	63.6	56.0	45.7	35.9
		MCDB	42.1	41.9	61.1	67.7	73.7	80.6	84.8	83.1	81.3	67.9	55.1	41.2
	5%	WB	34.4	34.3	44.1	48.4	58.6	64.4	68.6	66.1	60.7	52.0	41.3	33.6
		MCDB	38.2	38.0	54.9	62.2	72.1	78.0	81.8	82.4	76.0	63.2	48.7	37.2
	10%	WB	32.0	32.0	40.5	46.2	56.2	62.4	66.5	64.4	58.5	49.0	38.0	30.9
		MCDB	35.1	34.8	48.3	58.6	67.7	75.3	80.5	81.0	73.6	59.3	43.0	33.9
Mean Daily Temperature Range	5% DB	MDBR	21.7	21.7	22.6	26.6	26.8	26.3	29.8	30.4	30.3	26.4	23.6	21.4
		MCDBR	24.3	26.2	32.9	37.1	35.0	33.5	36.1	37.5	40.7	37.2	32.0	25.7
		MCWBR	17.5	18.5	19.4	20.2	17.0	14.7	12.9	14.3	17.6	19.3	21.1	18.9
	5% WB	MCDBR	22.0	23.2	32.3	34.5	29.6	28.5	29.1	31.0	33.7	33.9	31.5	23.9
		MCWBR	17.4	18.1	19.2	19.0	14.9	14.2	13.2	12.4	15.3	18.6	21.3	19.0
Clear-Sky Solar Irradiance	taub		0.250	0.269	0.275	0.319	0.338	0.351	0.359	0.367	0.323	0.287	0.259	0.243
	taud		2.472	2.463	2.543	2.425	2.424	2.430	2.425	2.377	2.469	2.560	2.535	2.502
	Ebn at Noon		270	288	304	296	293	288	284	277	281	278	265	260
	Edh at Noon		19	24	27	34	35	35	35	35	29	22	18	16
All-Sky Solar Radiation	RadAvg		465	790	1187	1567	1864	2037	2186	1789	1313	830	501	353
	RadStd		40	54	76	74	149	120	100	122	118	82	56	19

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page